

Mobility Lens – Assignment 1 Report

CS501 E1 — Mobile Application Development

Student Information

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Application package name: com.priyanshu.mobilitylens

Emulator

I ran the app on a Pixel XL emulator at API 33 (Android 13). The project compiles and targets 35. I installed that build on the API 33 image. compileSdk and targetSdk are what I built against; the emulator is what I actually clicked through for screenshots and the rotation test.

SDK Settings

minSdk is 24. I picked 24 so I was not stuck supporting really old phones, and Compose still works from that version.

targetSdk is 35 because that is the current release, and it is why I called enableEdgeToEdge() in MainActivity.

compileSdk is also 35. That is what the compiler builds against. The Pixel XL I tested on is API 33 and it ran fine. minSdk is what controls that, not compileSdk.

Key Project Files

MainActivity.kt is at

Assignment1-MobilityLens/app/src/main/java/com/student/mobilitylens/MainActivity.kt. It is only the launcher Activity. onCreate() calls setContent() and hosts MobilityLensApp(). I split the UI out of this file so it is not one giant composable.

MobilityLensApp.kt holds the remember state and lays out the screen. AppHeader.kt is the title bar. DimensionProgressBar.kt is the ticks and “Dimension X of 6.”

DimensionCard.kt is the name, description, and implication. NavigationButtons.kt is Previous and Next. AnalyzePanel.kt is the text field, Analyze, and the error or success message. DimensionCatalog.kt is the six dimension resource IDs, emojis, and colors. MobilityDimension.kt is the data class for those IDs.

AndroidManifest.xml is at Assignment1-MobilityLens/app/src/main/AndroidManifest.xml. Launcher activity, the theme, that is basically it. No extra permissions because the app never touches the network, camera, or location. I did not add INTERNET either. It has to be declared and I did not need it.

app/build.gradle.kts sets namespace and applicationId to com.priyanshu.mobilitylens, the three SDK levels, Java 11, and Compose. gradle/libs.versions.toml is the version catalog so I am not typing version numbers in five places.

strings.xml is at Assignment1-MobilityLens/app/src/main/res/values/strings.xml. The six dimension names, descriptions, implications, and the button labels are here instead of hardcoded in Kotlin.

The GitHub repo is CS501-E1. This homework is the Assignment1-MobilityLens folder. Screenshots are in Assignment1-MobilityLens/screenshots.

Rotation Observation

I was on Dimension 3, typed Snapchat in the text field, then rotated the emulator to landscape. The app landed back on Dimension 1 and the field was empty. I used remember for currentIndex and appNameInput. remember only holds state while the composable is alive. Rotation kills and recreates the Activity, so Compose starts fresh and both values go back to their defaults.

I did not fix it because the assignment said not to. But it matters on a real phone. Say someone is on Dimension 4, types their app name, then the phone rotates because they tilted it. Everything they typed is gone. rememberSaveable would write those values into saved instance state and restore them after the Activity comes back.

Mobile Design Decision

I gave each dimension its own accent color. The progress bar, the emoji circle, and the Next button all update together. The screen is small, so there is not much room for extra labels saying something changed. The color shift does that immediately. Even if someone cannot tell the colors apart, the emoji and the Dimension 2 label still update, so it is not color-only. Next is a large button in the lower half of the screen where a thumb can reach it. After Analyze I also hide the keyboard so the result is not stuck under it.